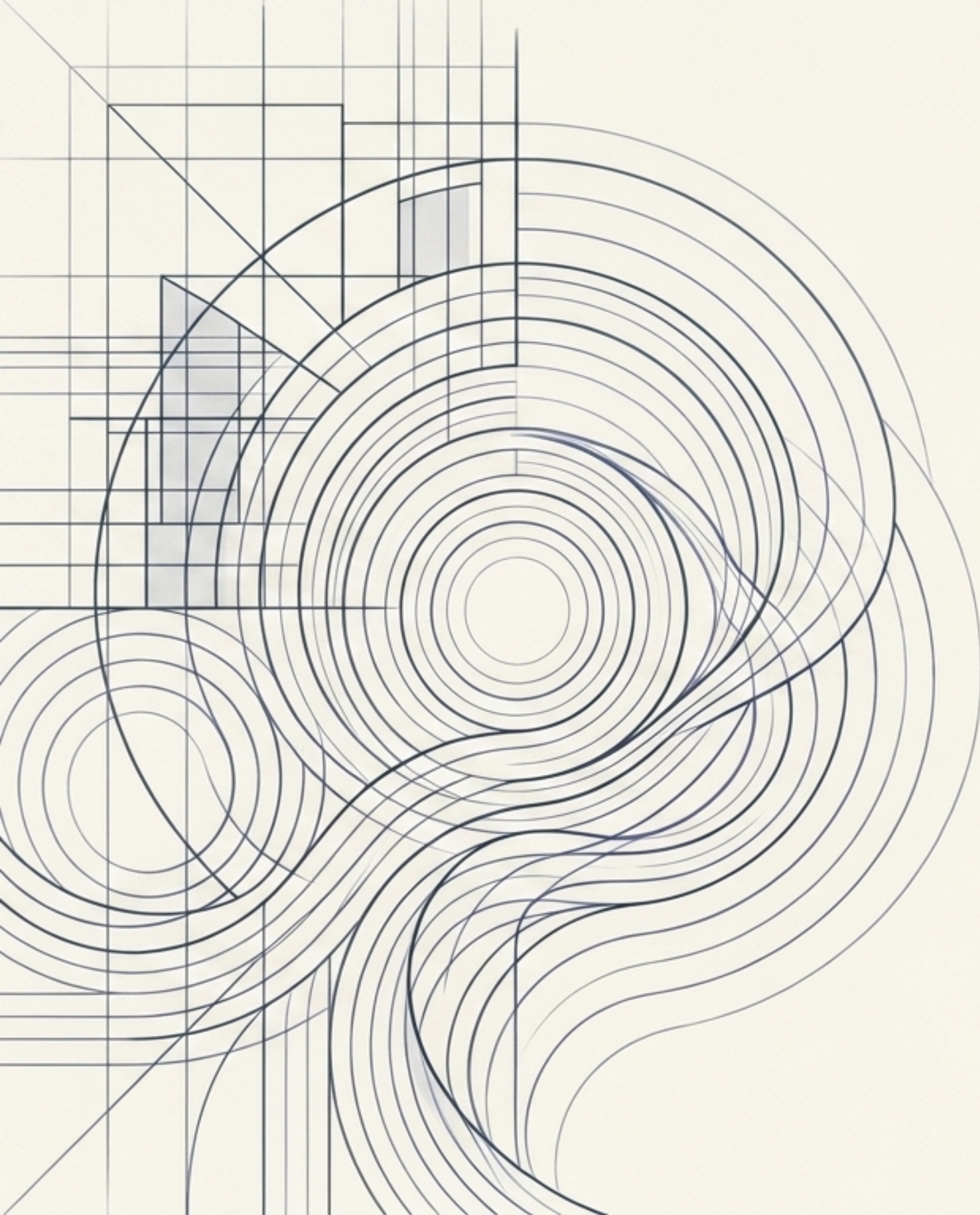
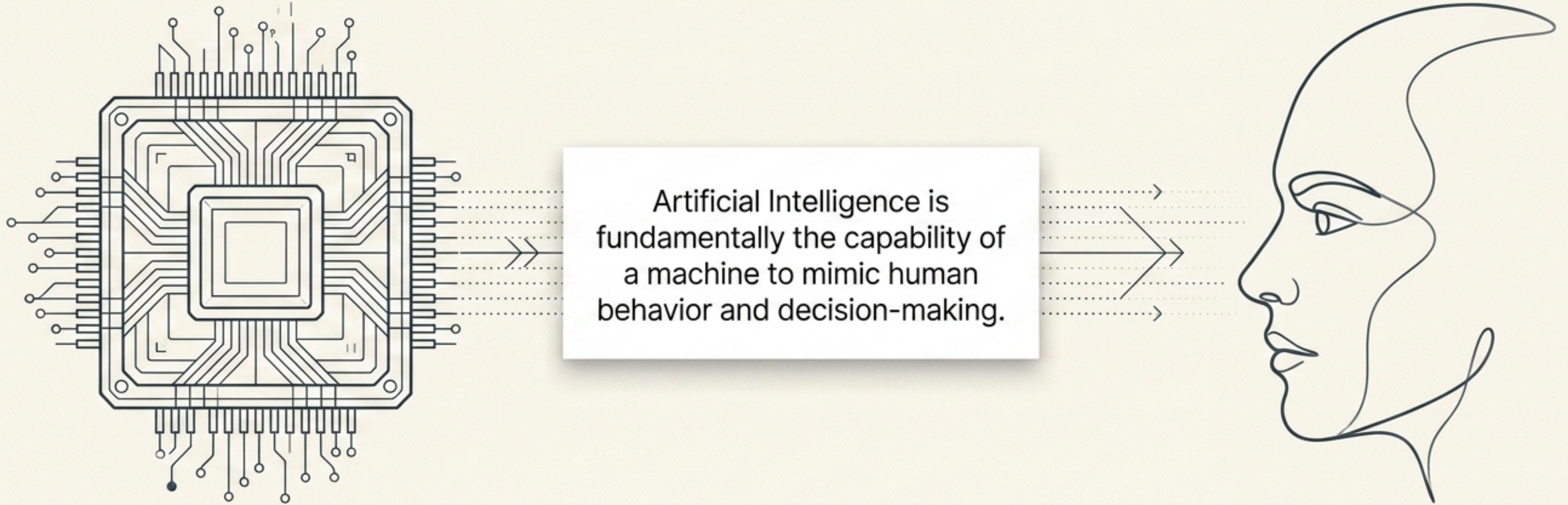




QA Gen AI & LLMs: Day 1 Agenda

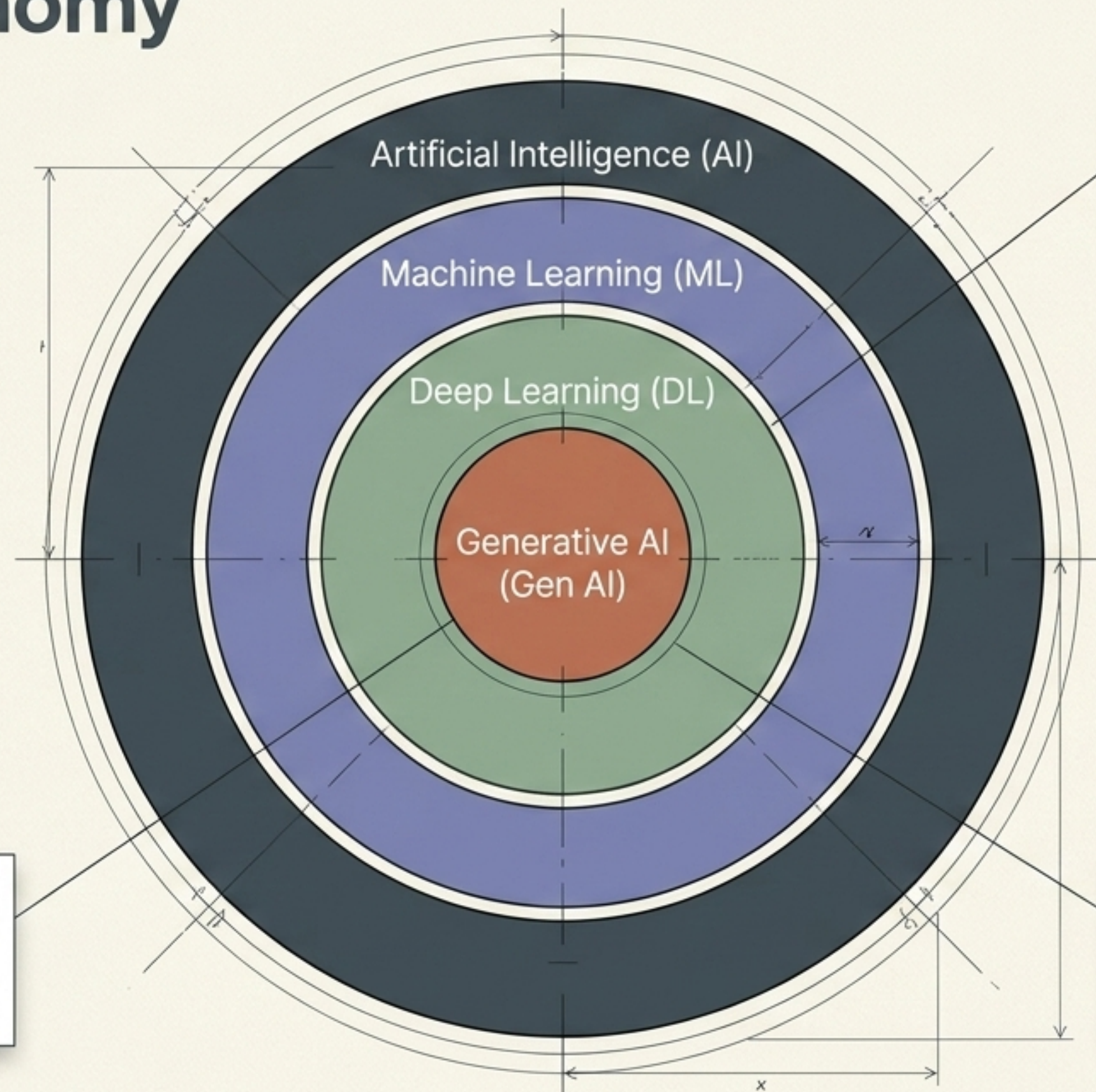
Demystifying the AI landscape for Quality Assurance.

The Evolution of AI: The Ultimate Goal



Every technology discussed today is simply a different method of achieving this mimicry.

The AI Taxonomy

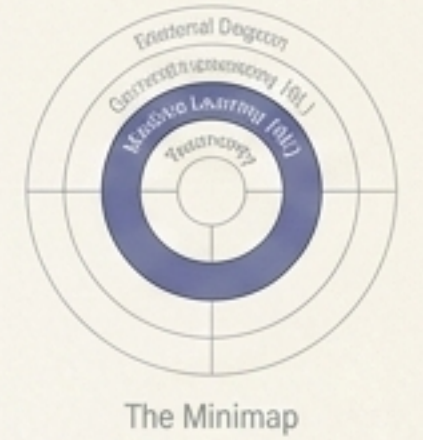


ML is the first sub-set of AI, focusing on prediction.

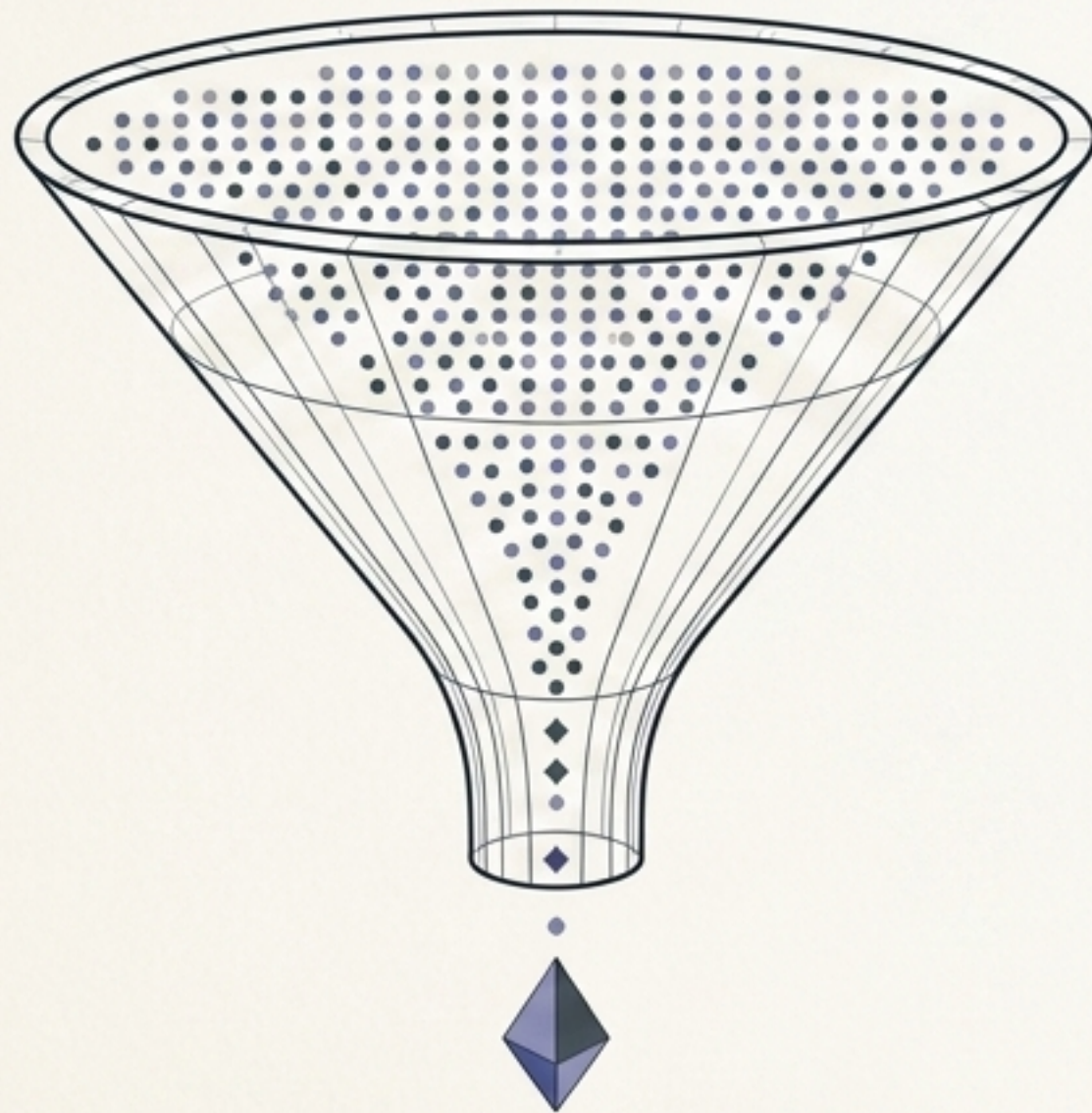
Gen AI is a subset of DL where the output shifts to generation.

DL is a specialized subset of ML using deep neural networks.

Machine Learning (ML): The Predictive Engine



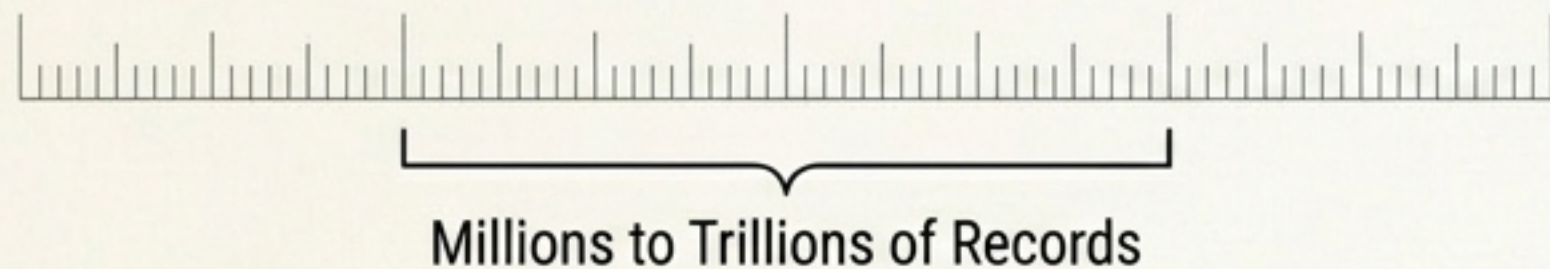
Data Scale: 1,000s to Millions of Records



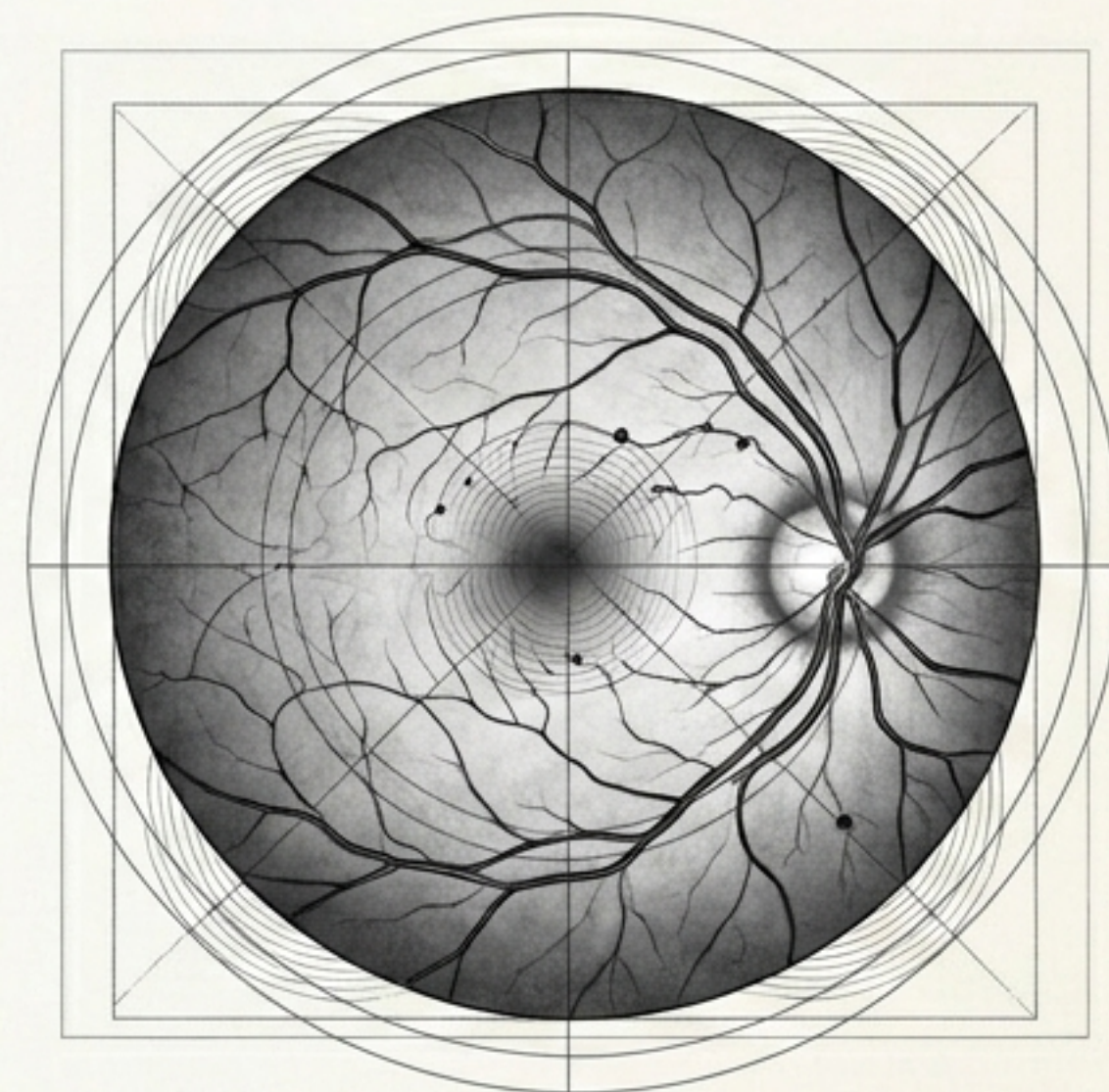
Predominantly Predictions

Supervised ML Types	
Linear Regression	Classification
Predicts numerical values.	Predicts discrete categories.
QA Use Case: Defect Prediction Extracting commit-based metadata from Git/Bitbucket to predict future defect counts to enable Risk-Based Testing.	Health Use Case: Heart Disease Predicting risk (Yes/No) based on historical patient records (Parameters: Name, Age, Blood group, Diabetes, BP, Genetic History).

Deep Learning (DL): The Human-Like Network



A specialized subset of ML utilizing deep, **multi-layered neural networks** to analyze vast amounts of unstructured data and think like a human.

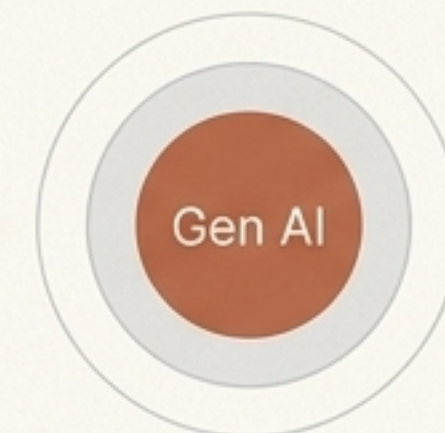


Convolutional Neural Network (CNN) Model

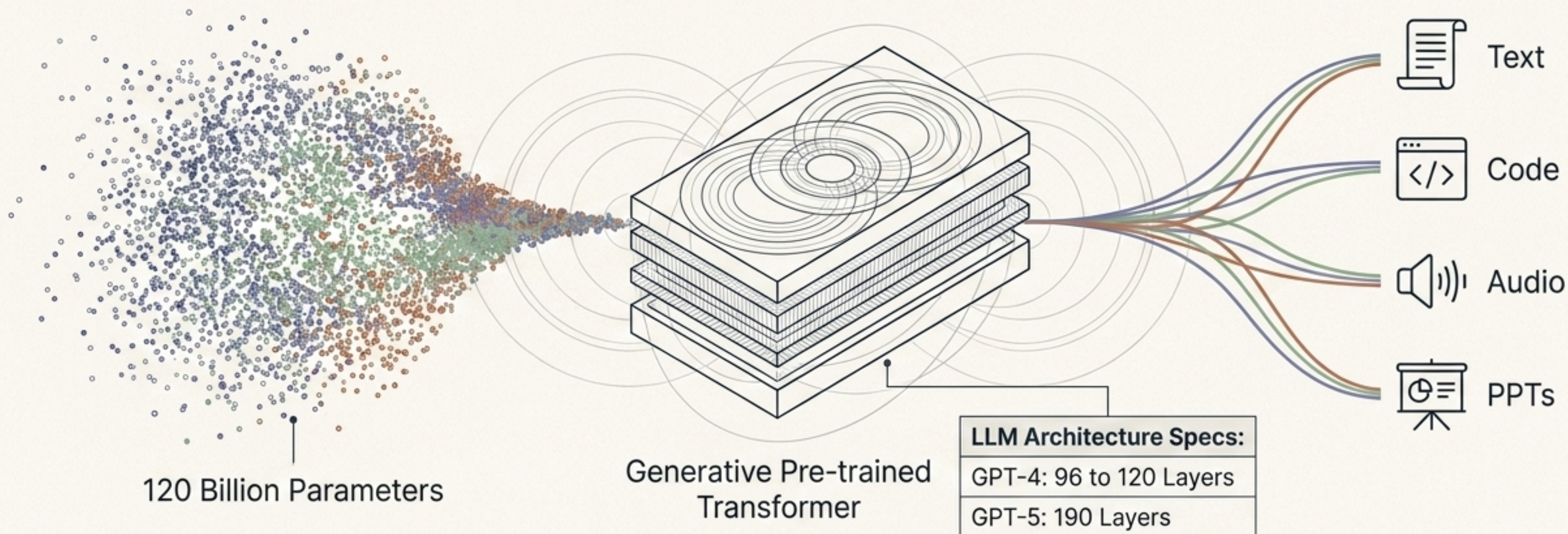
Use Case Spotlight: Analyzing unstructured visual data (retinal imaging) to predict health risks without explicit spreadsheet parameters.

Generative AI: The Creator

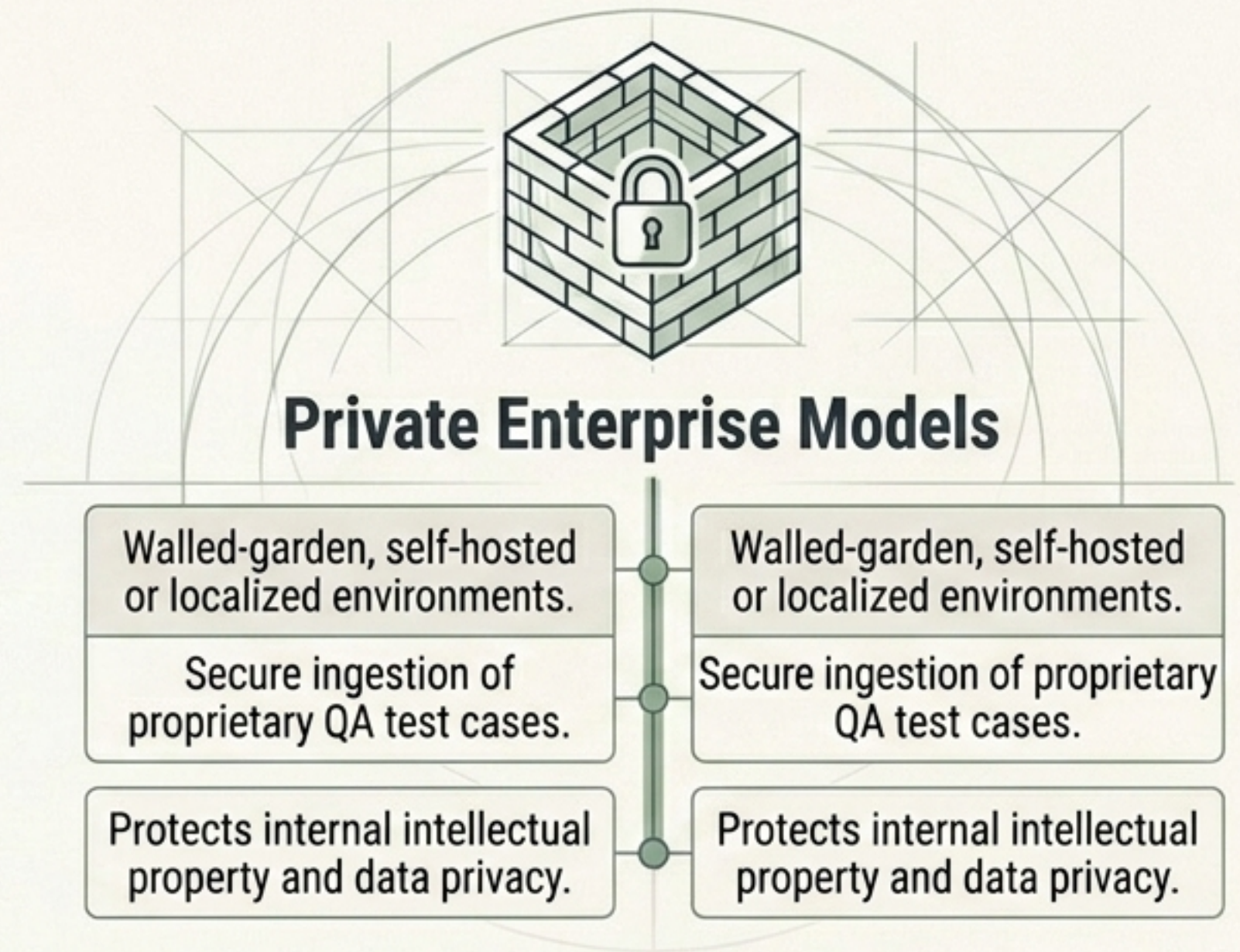
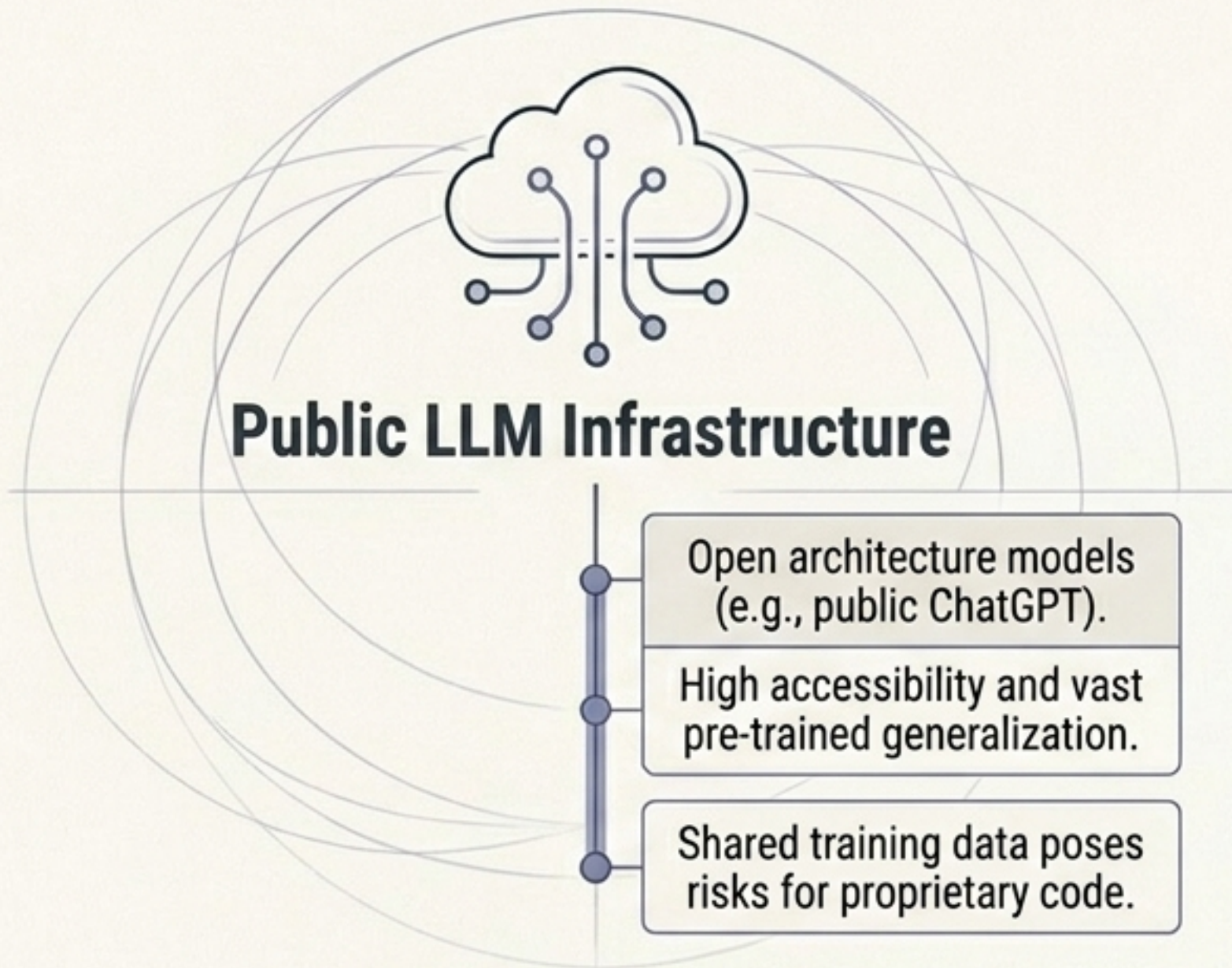
The Minimap



**The Paradigm Shift: Output is Generation, not Prediction.
Gen AI creates net-new content.**



The Path Forward: Public vs. Private LLMs



Discussion: As QA engineers, how do we balance the power of LLMs with the data privacy required for our proprietary codebases and test suites?